

```

1 c d
gotoxy(i,0); //Points the
cursor to location: lth row & 0th column
on the LCD.

while(j<(width-1))
{
    lcd_putc(pgm_read_byte(&string[j])); //
    Prints the character at the current cursor
    location.
}

j++;

} //WaitMs(RATE);

}

j=0; //Start first line
while(j<(len-1))
{
    lcd_clear();
    lcd_g_o -
}

toxy(0,0);
for(k=0,i=1;j<(k<16) && (pgm_read_
byte(&string[i])!='\0') ;k++,i++)
{
    lcd_putc(pgm_read_byte(&string[i]));

    WaitMs(RATE);
    j++;
}

return 0; //End of main
function.

}

void WaitMs(unsigned int ms) //
Generates a delay of ms milliseconds.
{
    unsigned int i;
    for(i=0;i<ms;i++)
    {
        _delay_ms(1);
    }
}

```

LCD2.H

```

#ifndef LCD_H
#define LCD_H

/*
 * Definitions for LCD controller type
 * Adapt the MCU clock frequency in Hz
 * to your target.
 */
#define XTAL 16000000 //MCU clock frequency in Hz, used to calculate
delay timer */

/* Definition for LCD controller type */
#define LCD_CONTROLLER_HD44780 0 //Use
0 for HD44780 controller, 1 for KS0073
controller */

/*
 * Definitions for Display Size
 * Change these definitions to adapt set-
 * ting to your display
 */
#define LCD_LINES 2 //number of visible lines of the display
#define LCD_DISP_LENGTH 16 //visible characters per line of the display
#define LCD_LINE_LENGTH 0x40 //internal line length of the display
#define LCD_START_LINE1 0x00 //DRAM address of first char of line 1
#define LCD_START_LINE2 0x40 //DRAM address of first char of line 2
#define LCD_START_LINE3 0x14 //DRAM address of first char of line 3
#define LCD_START_LINE4 0x54 //DRAM address of first char of line 4
#define LCD_WRAP_LINES 0 //0: no wrap, 1: wrap at end of visible
line */

#define LCD_IO_MODE 1 //memory mapped mode, 1: IO port mode */
#define LCD_PORT PORTC //port for the LCD lines
#define LCD_DATA0_PORT LCD_PORT //port for 4bit data bit 0
#define LCD_DATA1_PORT LCD_PORT //port for 4bit data bit 1
#define LCD_DATA2_PORT LCD_PORT //port for 4bit data bit 2
#define LCD_DATA3_PORT LCD_PORT //port for 4bit data bit 3
#define LCD_DATA0_PIN 3 //pin for 4bit data bit 0
#define LCD_DATA1_PIN 5 //pin for 4bit data bit 1
#define LCD_DATA2_PIN 6 //pin for 4bit data bit 2
#define LCD_DATA3_PIN 7 //pin for 4bit data bit 3

#define LCD_RS_PIN 2 //pin for RS line
#define LCD_E_PIN 3 //pin for Enable line
#define LCD_RW_PIN 1 //pin for RW line
#define LCD_E_PORT PORTC //port for Enable line
#define LCD_RS_PORT PORTD //port for RS line
#define LCD_RW_PORT PORTD //port for RW line

#define LCD_FUNCTION_2LINES 3 //DB3: two lines (0->56h line)
#define LCD_FUNCTION_1DOT5 2 //DB2: 5x10 font (0->5x7 font)
#define LCD_CURSOR 6 //DB6: set CG RAM address
#define LCD_CURSOR_ON 7 //DB7: set D5 RAM address
#define LCD_BUSY 7 //DB7: LCD is busy

/* set entry mode: display shift on/off,
dec/inc cursor move direction */
#define LCD_ENTRY_DEC 0x04 //display shift off, dec cursor move dir
#define LCD_ENTRY_DEC_SHIFT 0x05 //display shift on, dec cursor move dir
#define LCD_ENTRY_INC 0x06 //display shift off, inc cursor move dir
#define LCD_ENTRY_INC_SHIFT 0x07 //display shift on, inc cursor move dir

/* display on/off, cursor on/off, blinking
char at cursor position */
#define LCD_DISP_OFF 0x08 //display off
#define LCD_DISP_ON 0x0C //display on, cursor off
#define LCD_DISP_ON_BLINK 0x0D //display on, cursor off, blink char
#define LCD_DISP_ON_CURSOR 0x0E //display on, cursor on
#define LCD_DISP_ON_CURSOR_BLINK 0x0F //display on, cursor on, blink char

/* move cursor/shift display */
#define LCD_MOVE_CURSOR_LEFT 0x10 //move cursor left (decrement)
#define LCD_MOVE_CURSOR_RIGHT 0x14 //move cursor right (increment)
#define LCD_MOVE_DISP_LEFT 0x18 //shift display left
#define LCD_MOVE_DISP_RIGHT 0x1C //shift display right

/* function set: set interface data length
and number of display lines */
#define LCD_FUNCTION_4BIT_1LINE 0x20 //4-bit interface, single line, 5x7 dots
#define LCD_FUNCTION_4BIT_2LINES 0x28 //4-bit interface, dual line, 5x7 dots
#define LCD_MOVE_DEFAULT ((LCD_ENTRY_MODE) | (LCD_ENTRY_INC))
extern void lcd_init(uint8_t t, uint8_t x, uint8_t y);
extern void lcd_putc(uint8_t c);
extern void lcd_puts(const char *s);
extern void lcd_puts_p(const char *progmem_s);
extern void lcd_showvalue(unsigned char num);
extern void lcd_command(uint8_t cmd);
extern void lcd_data(uint8_t data);
#define lcd_puts_P(s) lcd_puts_P(PSTR(s))
#endif //LCD_H

```